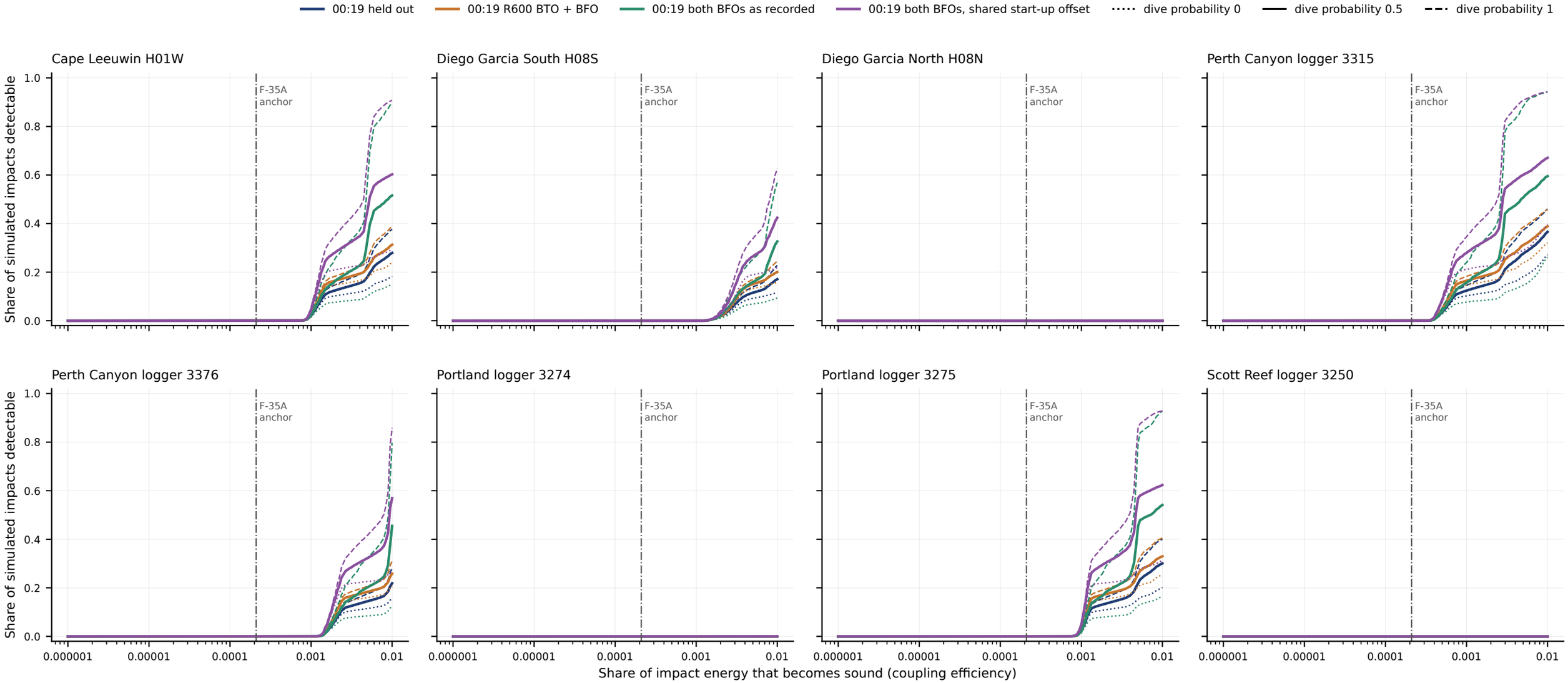
Share of simulated impacts detectable at each station against the coupling efficiency — baseline detector, STA/LTA (15 dB)



Plain-language note: Each line shows, for one 00:19 data option and one dive probability, the share of simulated impacts whose predicted 5–40 Hz level would exceed the detector threshold at the station, as the assumed share of impact energy that becomes sound varies over four decades. The dash-dot line marks the value calibrated on the F-35A crash, the upper anchor. A share near zero at the anchor means the station cannot be expected to have heard the impact.

Technical note: Source posterior: end of flight run J (74 seeds, 313.6 M rows) + northern supplement; weighted subsample 2 M draws per option x p; prereg run_j_levels.py 4277320. Threshold SNR >= 15 dB on the 5-40 Hz 1 s-equivalent total against the module's noise medians (model-matched upper bound from the weak-signal trial, 3376 background, alpha 0.01). Blocked paths count as undetectable. Source and TL as in the noise-versus-levels charts. PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, now a source-term caveat after the EW0112 result) and C2 (coupling efficiency not calibrated; F-35A upper anchor 2.1e-4).